Addressing Personalized Virtual Reality for Stroke Rehabilitation through a Country Fair Serious Game Approach: A proposal

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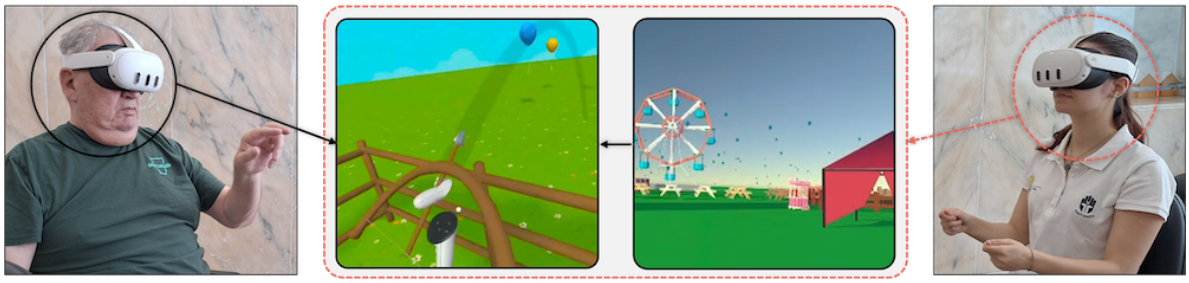


Figure 1: Stroke survivor experiencing a VR-based Country Fair experience, after it is personalized by a healthcare professional.

Abstract

Traditional rehabilitation for stroke survivors adopt uniform approaches, which can reduce motivation, engagement, and the effectiveness of recovery. This paper introduces a Personalized Virtual Reality (VR) experience for Stroke Rehabilitation through a Country Fair serious game approach. The rehabilitation takes the form of a virtual fairground composed of multiple tents, each hosting a distinct activity. These will be designed with customizable parameters, e.g., task difficulty, movement range, or cognitive demands, that can be adapted by healthcare professionals to match each survivor. The metaphor of a country fair also adds an engaging dimension. Survivors can explore diverse experiences that feel playful, varied, and never repetitive. This adaptability can foster sustained participation, increase the sense of agency, and align rehabilitation with the survivor's evolving progress. The paper defends the benefits of personalized VR in stroke recovery, outlines the potential of the Country Fair approach, and identifies directions for future research and development.

Keywords: Virtual Reality, Human-Centered, Personalization, Serious Game, Stroke, Rehabilitation, Adaptive Recovery

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1 Introduction

Stroke brings sudden and profound changes to the lives of survivors. Depending on the severity and location of the brain injury, individuals may face difficulties with mobility, balance, coordination, or speech, as well as challenges in memory and cognitive function. These impairments can severely limit independence and make even the simplest daily tasks, such as walking, eating, or dressing, difficult to perform without assistance [8, 24, 26].

For many survivors, the consequences extend beyond physical limitations. Stroke can affect social participation, self-esteem, and overall quality of life, as individuals struggle to adapt to new challenges in personal and professional settings. The combined impact of motor, cognitive, and emotional difficulties highlights the critical need for strategies that not only restore physical function but also support reintegration into everyday life [13, 15, 16].

In such a context, rehabilitation plays a central role in stroke recovery by helping survivors regain lost skills and relearn functions that were disrupted by brain injury. Through structured therapy, individuals can work to improve their mobility, coordination, and muscle strength, while also training their speech and cognitive abilities [3, 19, 21]. These interventions provide a pathway to greater independence, allowing survivors to progressively re-engage in essential daily activities. Ultimately, effective rehabilitation enhances the overall quality of life for stroke survivors, empowering them to rebuild their lives after a life-changing event [2, 22, 22].

Despite its critical role, traditional stroke rehabilitation methods often face some challenges. Many conventional exercises involve repetitive, monotonous movements that can feel tedious over time, leading to reduced motivation and disengagement. This lack of stimulation not only makes therapy less enjoyable but can also slow down recovery, as patients may be less likely to fully commit to their programs. In some cases, prolonged disengagement can even contribute to regression, undermining the progress already achieved [4, 18, 23].

Another limitation is the limited layer of personalization associated with traditional rehabilitation approaches. Many programs apply standardized exercises to all survivors, without sufficient adaptation to their unique abilities, needs, or pace of recovery. This

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one-size-fits-all approach may leave some of them overwhelmed by tasks that are too difficult, while others may feel under-challenged and unstimulated. Both scenarios reduce the overall effectiveness of rehabilitation, emphasizing the need for more adaptive, engaging, and individualized solutions [5, 10, 25, 27].

In this vein, this position paper proposes a Personalized Virtual Reality (VR) approach for stroke rehabilitation using a Country Fair serious game approach. Using VR enables the creation of highly controlled, varied, and stimulating rehabilitation scenarios, allowing for the simulation of real-world tasks that encourage active participation. Moreover, it provides immediate feedback, helping stroke survivors track their progress and improve motivation over time [11, 14, 20]. By combining VR with personalization methods, it may be possible to address the varying levels of ability among survivors, ensuring that rehabilitation programs are appropriately challenging yet achievable [1, 12, 28].

Our vision focuses on a rehabilitation environment composed of a virtual fairground with multiple tents, each offering a distinct, customizable activity. These activities function as serious games with adjustable parameters (task difficulty, movement range, or cognitive demands), allowing healthcare professionals to tailor challenges to each survivor's abilities. The fairground metaphor adds an engaging, playful dimension, encouraging exploration and variety while preventing repetition. By combining personalization with enjoyable, achievable challenges, this approach aims to enhance motivation, foster sustained participation, and empower survivors as their abilities evolve. These and other arguments in favor of this vision are presented, alongside future directions for research and development.

2 Methods and Materials

This line of research began with a scoping review of personalized immersive experiences for stroke rehabilitation within the SCOPUS database, covering the period from 2020 to 2024. The review highlighted current trends, advancements, and challenges in the field, while also pointing to opportunities for future development [8].

The findings revealed that most existing VR-based rehabilitation solutions are developed with limited involvement of the target audience. Although some serious games exploring personalization methods are emerging, they are often driven by technological feasibility rather than human-centered requirements, a limitation further reinforced by the general lack of user evaluation in this domain [17].

To address this gap and building on a long-standing collaboration with a renowned rehabilitation center in Portugal, the research team decided to adopt a Human-Centered Design (HCD) approach before moving into the creation of personalized serious games. This process began by gathering feedback from stroke survivors and healthcare professionals at the center, regarding their needs, preferences, and perspectives on personalized VR rehabilitation (Figure 2).

A tailored questionnaire was developed for each group, including demographic questions, multimedia explanations of VR and its potential benefits for personalization, and prompts about preferred activities and adjustable features. For healthcare professionals, additional questions focused on the timing, frequency, and platforms for personalization, as well as other relevant clinical considerations [6].

The collected data informed a series of brainstorming sessions, where the team discussed insights, identified patterns, and iteratively defined requirements to guide the next stages of development. From this process, a conceptual model was formulated to articulate the vision of supporting stroke survivors through Personalized VR-based Rehabilitation [7, 9].

2.1 Conceptual model for VR-based Personalization

As illustrated in Figure 1, our proposal focuses on a Personalized VR-based country fair, composed by a set of distinct serious games, offering a novel and engaging framework for stroke recovery. In this model, rehabilitation is re-imagined as a virtual fairground, where



Figure 2: Research team at a rehabilitation center, conducting a data collection session, through questionnaires and interviews, gathering information on stroke survivors' preferences regarding the use of Personalized VR for assisting with the rehabilitation process.

different tents and booths represent distinct therapeutic activities. Each tent is designed to evoke the atmosphere of a traditional country fair attraction, yet beneath this playful metaphor lie exercises carefully aligned with the goals of stroke rehabilitation. Survivors (equipped with a VR headset) are invited to step into these virtual tents and perform meaningful actions that simultaneously promote physical recovery and stimulate cognitive functions. By transforming the clinical setting into a socially resonant and familiar experience, we strongly believe that the fair metaphor can help increase motivation, engagement, and adherence.

Importantly, this playful metaphor is not merely cosmetic. The use of a country fair evokes a sense of joy, familiarity, and nostalgia, counteracting the monotony often associated with traditional rehabilitation. Survivors are more likely to remain engaged in their therapy when the activities feel meaningful and enjoyable. At the same time, the fair structure is highly scalable: new tents can be added as new therapeutic modules without disrupting the overall metaphor, and the framework can be applied equally well in centers or in home-based rehabilitation.

The fairground serves as a metaphor for a personalized rehabilitation environment. Prior to the rehabilitation sessions, healthcare professionals (either in a computer or in the VR headset as well) can tailor the fair to each individual survivor, selecting which tents appear and in what order they should be visited. In this way, the rehabilitation journey becomes a pathway through the fair, one that can be adjusted to the survivor's specific motor, cognitive, or perceptual needs. For example, a survivor with fine motor difficulties may start with games focused on precision and grip strength, like tossing rings onto pegs or picking up small objects at a prize booth.

A defining feature of this model is its adaptability at multiple levels. Each serious game inside a tent can be finely tuned to match the survivor's abilities. In a ball-toss booth, for instance, parameters can be adjusted by changing the number and size of the cups, altering the physics of the ball, or varying the distance to the targets. Early sessions may begin with larger targets placed close at hand, while later sessions increase the challenge by making targets smaller and farther away. This adaptability is not static; rather, the proposed approach can store distinct metrics regarding the survivor's performance (e.g., accuracy, speed, movement smoothness, others), and use the data from previous sessions to help inform the healthcare team. These data can then be used to personalize the experience,

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ensuring the survivor is always working within an optimal zone of difficulty, neither too easy nor overwhelmingly hard.

The personalization layer extends beyond single tasks and governs the overall rehabilitation journey. Professionals can design a fairground path tailored to each survivor, combining different tents into a coherent program. As survivors progress, new tents can be unlocked, representing higher levels of challenge or different rehabilitation domains, such as memory games, problem-solving puzzles, or cooperative activities designed to support emotional well-being and social reintegration. In this way, the fairground evolves alongside the survivor, providing both a visual representation of progress and a motivational reward system.

In short, the country fair approach brings together a set of distinct benefits for stroke rehabilitation. It can transform repetitive exercises into playful challenges, increasing motivation and engagement. It may also support holistic rehabilitation by combining motor, cognitive, and emotional domains within a single coherent environment. Its modular design allows scalability, with new tents easily added as rehabilitation evolves. Moreover, its accessibility is also a key advantage, as survivors can explore the fair metaphor in any place as long as they have the VR equipment. Finally, the metaphor resonates emotionally by evoking joy, nostalgia, and a sense of community, reducing therapy fatigue and encouraging long-term participation.

3 Arguments in favor of our proposal

Next, various arguments to support our vision are described.

- **Engagement through a familiar and playful metaphor:** The country fair metaphor (tents, games, colorful setting) can transform rehabilitation into a fun, familiar, and motivating experience. Gamified environments reduce the perception of repetitive, tedious exercise by framing them as a fair activity. Increased engagement can lead to higher adherence to rehab routines, which is critical for recovery outcomes;
- **Cultural resonance and emotional value:** A country fair evokes nostalgia, joy, and community, which can boost mood and emotional well-being, important for post-stroke recovery, where depression and demotivation are common. Anchoring rehab in a positive cultural image may reduce stigma associated with therapy and increase willingness to engage;
- **Narrative & sense of progression** Unlike static exercises, the fairground offers a narrative of exploration: moving from tent to tent, unlocking new activities, or earning symbolic rewards. This sense of story and progression can increase commitment, sustain long-term engagement, and help survivors feel they are on a meaningful journey rather than repeating static activities;
- **Rewards & unlockable progression** A reward system, where survivors earn tokens or prizes by completing activities, can be used to unlock new tents, or fair zones. This gamified progression reinforces motivation and provides tangible feedback on improvement. Over time, the rewards also serve as markers of progress, giving survivors a visible sense of achievement that extends beyond individual sessions;
- **Personalization and flexibility:** Each tent represents a modular activity. Healthcare professionals can design a custom path through the fair, tailoring the sequence and difficulty of activities to each survivor's needs. Distinct parameters can be personalized (speed, distance, accuracy, number of attempts) to keep the serious games level optimal for recovery progression;
- **Multimodal rehabilitation:** Fair activities can combine gross motor tasks (throwing balls, pulling levers, etc.), fine motor skills (stacking, aiming, etc.), and cognitive challenges (memory games, sequencing, strategy). This way, stroke survivors can benefit from environments that stimulate multiple domains simultaneously, reflecting real-world complexity;
- **Progress tracking and data-driven adaptation:** Each session generates performance metrics (reaction time, success rate, movement smoothness, endurance). The proposed concept can adapt future rehabilitation sessions based on survivors' performance, and professionals' insight, creating a mechanism for continuous improvement;
- **Scalability and inclusivity** A fairground metaphor scales easily: new tents = new rehab modules, without disrupting the overall framework. The modular tent system makes it possible to integrate activities for survivors with diverse profiles, including those with predominant cognitive deficits, language impairments, or social anxiety. By including a broad range of tents, the fair becomes an inclusive rehabilitation environment where every survivor can find activities suited to their needs;
- **Distinct roles** The fair can enable dual interaction: survivors may participate in the various serious games, while healthcare professionals can explore the fair alongside them in guiding roles. This supports real-time monitoring and encouragement, as well as the capacity to take notes for post-processing and adjustment over the next rehabilitation sessions;
- **Collaborative and competitive settings** The fair metaphor evokes social participation. It can support multiplayer modes where survivors engage in cooperative challenges (e.g., team ring toss) or friendly competition (e.g., dart-throwing contest). These settings foster social interaction, peer support, and playful rivalry, which can increase engagement;
- **Community and remote extension** The proposed metaphor can extend beyond rehabilitation centers. Survivors will be able to access the fair remotely, maintaining continuity of therapy at home while still feeling part of a community of peers and professionals. This can foster long-term engagement and social connection, even outside formal rehabilitation sessions.

4 Framework overview

Figure 3 illustrates a framework for implementing the VR-based personalized country fair approach. The stroke survivor is the primary user of the framework, engaging directly with the virtual rehabilitation activities based on serious games designed to support motor, and cognitive recovery. Wearing a VR headset (e.g., META QUEST 3), the survivor can visualize the immersive VR environment and interact naturally with its elements, as the device's cameras and sensors capture and track hand movements in real time.

The **VR environment** represents the immersive country-fair world where rehabilitation activities take place. It contains the various tents, and navigation paths that form the survivor's personalized therapy experience. This environment captures detailed interaction data, such as survivors movement accuracy, reaction speed, and task success, generating data that feed the adaptive personalization process which are then forwarded to the remaining modules.

The **Data Hub** is the analytical core of the framework. It aggregates the information collected during gameplay and processes it to extract meaningful indicators of performance and progress. By comparing current results with previous sessions stored in its database, the data hub identifies trends, difficulties, and improvements. These insights form the basis for personalized adjustments in future rehabilitation sessions.

The **Personalization Module** uses the insights generated by the data hub to adapt game parameters and tailor each activity to the survivor's evolving needs. This can include adjusting difficulty, target size, distance, speed, or the type of feedback provided. The updated

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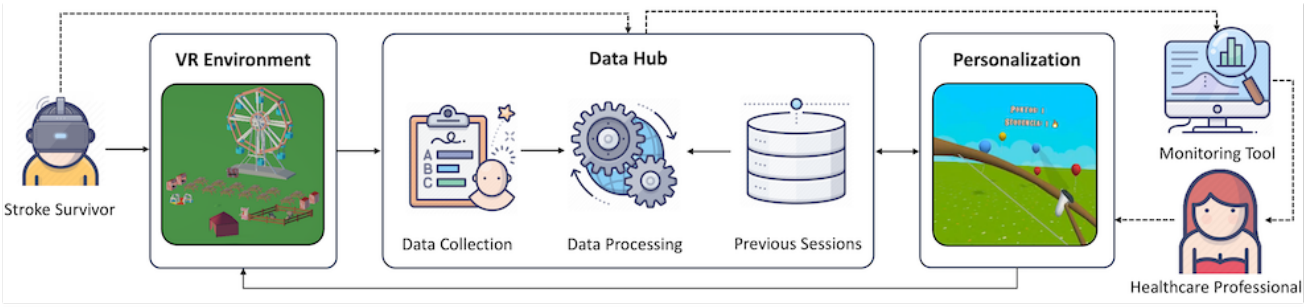


Figure 3: Framework for using a personalized VR-based country fair approach for stroke survivors' recovery. During game-play, data collection is performed, using distinct metrics from the survivor and the status of the VR environment. This data feeds the data hub, which can also access data from previous sessions and feeds the monitoring tool for the healthcare professional to access and make informed decision. Afterwards, the experience will be adjusted for each survivor. Assets from iconfinder.com

configurations are then applied to the next rehabilitation session, ensuring that rehabilitation remains appropriately challenging, motivating, and suited to the survivor's abilities. The personalization process can be carried out automatically by the framework, which adjusts game parameters based on performance data gathered across sessions, or it can be manually overridden by healthcare professionals who review this information through a dedicated **Monitoring tool**. This dual approach ensures that the rehabilitation experience remains both adaptive to the survivor's evolving abilities and aligned with clinical judgment when specific therapeutic adjustments are needed. Regarding the monitoring interface, it allows healthcare professionals to review performance data and session summaries in an accessible way. They also receive feedback about how survivors are interacting with the system, enabling more informed decision-making and individualized rehabilitation planning.

All in all, this scalable proposal makes it possible to integrate a wide range of VR-based serious games within the country fair theme, offering healthcare professionals a flexible pool of options to tailor each survivor's rehabilitation journey. By allowing activities to evolve over time and adapt to individual needs, the framework maximizes the therapeutic potential of every game while avoiding the limitations of a one-size-fits-all approach to rehabilitation.

4.1 Example of VR-based Personalized Serious Game

To illustrate how a VR-based experience can align with the proposed vision of personalization for stroke rehabilitation, we first describe a serious game inspired by an archery setting. Afterwards, other experiences are introduced, following similar principles of personalization, but described with less detail.

As a survivor enters the archery tent, they are greeted by a virtual avatar, the fair host, who introduces them to the rehabilitation activity and sets the tone for the experience. The host explains the purpose of the game, outlines the specific goals to work toward, and highlights the rewards that can be earned by completing challenges. This introduction functions as a brief tutorial, guiding survivors through the essential mechanics and helping them feel comfortable with the interaction patterns before beginning the activity. This helps promote survivors' confidence as they progress through the activity.

Moving forward, survivors will engage in virtual archery challenges that require controlled arm movements and sustained focus—skills closely tied to both motor and cognitive recovery. The activity begins gently, with a single, large bullseye target positioned at a short distance to help survivors become comfortable with the motion and rhythm of drawing and releasing the bow. As they progress, the distance to the target gradually increases, the number of targets begins to vary, and the target size becomes smaller. In advanced rounds, moving targets may be introduced (e.g., pop colorful balloons instead of traditional bullseye targets), adding a new layer of complexity that demands improved coordination, attention, and timing. Throughout this progression, survivors earn rewards for completing levels or demonstrating improved performance, which can unlock new features such as decorative bow attachments, the ability to shoot multiple arrows at once, or playful effects like flaming arrows. These customizable elements enrich the experience with enjoyment and novelty, helping maintain interest and motivation as survivors continue through their rehabilitation journey.

This activity supports rehabilitation by strengthening the upper limb, improving coordination, and enhancing postural control, while also engaging cognitive skills such as focus, planning, and timing. The act of drawing and aiming the bow promotes controlled arm movements and stability, while aligning each shot with a target encourages concentration and precise motor coordination. Adjusting distance, size, or movement of targets further challenges survivors to plan their actions and time their releases effectively, blending physical effort with cognitive engagement in a simple, motivating way. Survivors may initially perform the activity using only their non-affected arm to build confidence and learn the basic mechanics, then gradually transition to using their affected arm as strength and control improve, and ultimately progress to engaging both arms together, supporting more balanced motor recovery and encouraging coordinated bilateral movement.

From a personalization point of view, even a simple activity like archery can be finely tuned by healthcare professionals to match each survivor's abilities and progress. Target size, number, and type can be adjusted to increase or reduce precision demands, while the distance to the targets or the introduction of moving elements allows the task to evolve in difficulty over time. These adjustable parameters ensure that the experience remains appropriately challenging, supports gradual progression, and is tailored to the survivor's specific rehabilitation needs.

In addition, visual and auditory feedback should complement the experience by guiding, motivating, and supporting the survivor during the activity. On-screen cues or simple animations can signal progress or highlight areas for adjustment, while spoken or sound-based feedback can offer congratulations, encouragement, or reminders on how to perform specific movements. Together, these elements help maintain engagement, provide clarity on performance, and reinforce proper upper-limb movement patterns, contributing to a more effective and motivating rehabilitation experience.

All in all, this example demonstrates how immersive and adaptable VR experiences can transform rehabilitation into a more dynamic, engaging, and personalized process, offering survivors activities that evolve with their abilities while maintaining motivation and supporting meaningful progress.

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4.2 Other games for VR-personalized rehabilitation

Next, some additional examples of serious games that align well with the proposed country fair concept are presented, illustrating how they can be meaningfully integrated into the rehabilitation framework. By embedding these personalized games within the fairground metaphor, each one not only contributes to the rehabilitation process but also reinforces the vision of a dynamic, adaptable, scalable, and motivating environment for stroke recovery:

- **Ball toss - Can knockdown:** This game invites survivors to throw balls at stacked cans, a playful test of aim and coordination. Difficulty can be tuned by adjusting the ball's weight, the stability of the targets, or the throwing distance. The task strengthens grip, throwing motion, and precision while providing the joy of watching towers topple;
- **Ring toss:** In this game, survivors will practice precision and timing by tossing rings onto pegs. Professionals can vary peg distance, height, or even introduce moving targets to adjust challenge levels;
- **Whac-a-mole:** Survivors must hit virtual moles as they appear in unpredictable locations. Personalization involves adjusting the number of holes, the pop-up speed, or rules about which critters to hit. This game can help train reaction time, cognitive flexibility, and coordination in an engaging way;
- **Shooting duck gallery:** The duck shooting game provides a moving-target challenge, where survivors aim and fire at ducks gliding along a track. Target size, speed, and path variability can be adapted to suit ability levels. The game can sharpen visual tracking, focus, and upper-limb control, wrapped in the classic fairground atmosphere;
- **Fruit picking:** Survivors must collect fruit from trees or baskets, practicing reach and grasp in dynamic ways. The professionals can adjust fruit size, placement height, or whether fruit is static or falling. This activity supports flexibility, shoulder mobility, and coordination, all within a familiar and motivating scenario;
- **Skee-ball:** This activity replicates the classic of rolling balls up a ramp into scoring holes. Personalization options can include ramp incline, ball weight, or target hole size. Survivors may practice grip, release, and trajectory planning while enjoying the playful element of scoring points;
- **Basket toss:** Players toss balls into baskets placed at different distances, practicing controlled arm movements and coordination. The professional can personalize the basket height and size, the weight of the balls, and the number or timing of throws. This activity supports the development of arm control and coordination.
- **Fishing pond:** At this game, survivors use a rod or net to catch fish from a pond. The activity can be adjusted by altering fish size, movement speed, or rod/net sensitivity. The game can develop grip control, sustained attention, and fine motor precision in a calming, rewarding setting;
- **Memory match:** This game challenges survivors to remember and match pairs of objects, either images, numbers, symbols or others. Difficulty can be scaled by increasing the number of items, reducing exposure time, or adding distractions. The game can sharpen working memory, attention, and recognition while keeping the interaction playful;
- **Puzzle assembly:** Survivors are invited to solve puzzles by fitting pieces together or arranging shapes into patterns. The professionals can alter the piece number, shape complexity, or integrate familiar images into the puzzles. The activity may stimulate problem-solving and visual reasoning, with a clear sense of achievement at completion;
- **Treasure hunt:** Exploring the fairground, survivors will follow clues to locate hidden objects in sequence. The professionals can expand the search area, increase clue number, or vary its difficulty. This activity engages planning, attention, and navigation, blending cognitive training with the thrill of discovery;
- **Co-op challenge:** In this game, survivors can work with peers or professionals to complete a joint task, such as carrying a large object or performing a puzzle assembly. Parameters like object size, weight, and timing can be adapted to participant abilities. The activity fosters collaboration, communication, and motor coordination;
- **Friendly Competition:** Classic competitive fair games like dart throwing or ring toss can be played in head-to-head mode. Difficulty levels and scoring systems can be adjusted to ensure fairness and inclusivity. Competition can encourage motivation, social interaction, and resilience through playful rivalry;
- **Prize wheel:** Finally, survivors can spend tokens earned from other tents to spin a prize wheel. The professional may control token thresholds, prize types, or unlockable zones. This gamified reward system can reinforce effort, track progress, and sustain long-term engagement by giving survivors a sense of progression and achievement.
- **High Striker:** Survivors can swing a virtual mallet to strike a target and try to send a puck up a tower to ring the bell. The professional can personalize the required swing force, the sensitivity of motion detection, and the feedback intensity. This activity promotes upper limb strength, coordination, timing, and bilateral engagement while providing immediate feedback.
- **Frisbee Toss:** Survivors throw virtual frisbees toward moving targets around the fairground. The professional can personalize the game by adjusting the target's distance, size, and movement speed to suit each survivor's ability. This activity can promote arm strength, range of motion, and precision.

5 Future directions: a proposal

Various topics need to be addressed by the research community with the goal of exploring the impact of using a VR-based Country Fair approach for supporting personalized experiences for stroke rehabilitation, in particular:

- **Game design and integration:** Developing a playground that integrates multiple serious games under the fairground metaphor, ensuring smooth navigation, consistent interaction mechanics, and an engaging gameplay;
- **Narrative & sense of progression:** Exploring how storytelling elements, such as progression through fairground zones, unlocking new tents, or earning symbolic milestones, can sustain engagement and provide survivors with a clear sense of journey and achievement in their rehabilitation;
- **Selection/adaptation of personalization methods:** Selection/Designing adaptive methods (rule-based; bio-feedback; behavioral tracking; contextual adaptation; intelligent monitoring; others) to tailor game parameters based on the characteristics of each stroke survivor;

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- **Accessibility and inclusivity:** Ensuring the design accommodates survivors with varying physical, cognitive impairments, reducing barriers to use and maximizing inclusivity;
- **Scalability:** Designing the framework to support the seamless addition of new serious games and modules over time, enabling continuous evolution of the fairground and personalization of experiences as technology and therapeutic insights advance;
- **Motivation and gamification:** Including reward systems (tokens, prizes, unlocking new zones) to reinforce motivation, sustain participation, and provide survivors with a tangible sense of progress;
- **Professional-survivor interaction:** Defining ways for healthcare professionals to interact with the survivor and the game environment, monitoring progress, and adjusting activities in real time, if necessary when waiting for the next session is not the best course of action;
- **Data Collection and personalization:** Establishing mechanisms to capture distinct metrics (e.g., accuracy, movement quality, reaction times) and translating them into suggestions for healthcare professionals to make informed decisions and personalized next rehabilitation sessions;
- **Field validation:** Conducting trials in rehabilitation centers with diverse populations of stroke survivors to assess clinical outcomes, engagement, and real-world impact of the approach;
- **Longitudinal impact:** Studying the effects of sustained engagement with the Country Fair VR approach over extended periods, evaluating not only functional recovery but also quality of life and psychosocial outcomes;
- **Collaborative & competitive settings:** Investigating the role of multi-user features that allow survivors to collaborate or compete, fostering social interaction, teamwork, and peer motivation;
- **Remote rehabilitation & community building:** Examining how the VR-based fairground can extend beyond clinical settings into home environments, maintaining a sense of community and continuity of therapy through remote participation.

In addressing these topics, the field can advance towards the creation of truly personalized, adaptive VR rehabilitation experiences that can have a positive impact on the rehabilitation process of stroke survivors, as well as enhance how healthcare professionals can accompany this recovery process.

6 Concluding Remarks

Activities central to stroke recovery may be too difficult for some individuals due to their limitations, while others may find those same tasks less challenging. In this vein, the authors strongly believe that a valuable opportunity to explore the integration of adaptive tools in stroke rehabilitation exists, which can significantly enhance the personalization and effectiveness of recovery programs.

This work proposes the use of Personalized VR for Stroke Rehabilitation through a Country Fair serious game approach, having a virtual fairground composed of multiple tents, each hosting a distinct activity with customizable parameters, that can be adapted by healthcare professionals to match the survivor's specific abilities. Survivors can explore a diverse set of experiences that feel playful, varied, and never repetitive, thanks to the personalization layer. This dynamic adaptation could ensure that the rehabilitation program remains appropriately challenging yet attainable for every survivor. The goal is to promote larger engagement in the rehabilitation process, increasing adherence to therapy, and improving recovery rates.

Moving forward, to make this vision a reality, several important steps remain to be addressed, highlighting the need for interdisciplinary research. As an initial stage, some of the proposed games must be designed and integrated into a unified playground, with each activity represented by its own tent and supported by essential dynamics such as narrative, navigation, manipulation, and interaction. Building on this foundation, future work should focus on designing and validating adaptive methods capable of collecting and processing performance data to generate personalized suggestions for healthcare professionals, enabling them to make informed decisions and fine-tune game parameters. Equally critical will be the evaluation of these tools in real rehabilitation settings, engaging stroke survivors over extended periods to assess both clinical outcomes and user engagement. Together, these efforts are essential to fully realize the promise of adaptive, personalized VR as a transformative approach to stroke rehabilitation.

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